# Linear Growth Embedding of MAAT Structural Selection

Perturbation Stability, Positivity, and Sub-Percent Growth Signatures

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**Abstract**

Paper 34 showed that the MAAT projection observable does not reduce to the standard growth observable $f\sigma_8(z)$ under a one-parameter rescaling. The next step is therefore not another direct template comparison, but a forward embedding into a growth equation.

This paper introduces a minimal linear-growth embedding of MAAT structural selection. The matter growth equation is modified by a small effective MAAT source term proportional to the MAAT background fraction. In parallel, local selection-pressure perturbations $\delta\lambda_a(k,t)$ are evolved in Fourier space with diffusion and relaxation, and the positivity of $\lambda_a(x,t)$ is stress-tested numerically.

For the representative MAAT branch $\Omega_{\mathrm{MAAT},0} = 0.00331$ and $w_{\mathrm{MAAT}} = -0.801$, the corrected comparison to a true ΛCDM reference gives sub-percent deviations:

$$\max_{z<2} |\Delta D/D| = 0.0591\%, \qquad \max_{z<2} |\Delta f\sigma_8/f\sigma_8| = 0.4570\%.$$

The selection-pressure modes are linearly stable, all tested positivity scenarios remain non-negative, and the effective dark-energy-like branch satisfies the null, weak, and dominant energy conditions while violating the strong energy condition, as expected for accelerated expansion.

The result is not a precision cosmological analysis. It is a first reproducible forward-model step showing that MAAT structural selection can be embedded into linear growth without producing large or unstable perturbative effects.

## 1 Purpose and Status

Paper 34 established that the projection observable $C_{\mathrm{proj}}$ is not a growth observable by itself. Its mismatch with $f\sigma_8$ was interpreted as evidence that it probes a different structural information layer.

The present paper addresses the next question:

> *Can MAAT structural selection be embedded into a growth equation without destroying perturbative stability?*

The answer in this minimal benchmark is yes. However, the result is limited. The model is not a Boltzmann-code calculation, not a parameter inference, and not a joint cosmological constraint. It is a controlled perturbative toy embedding designed to test whether the MAAT branch produces small, stable, and measurable growth-sector signatures.

## 2 Background Model

The reference cosmology uses

$$H_0 = 67.4, \qquad \Omega_{m0} = 0.315, \qquad \Omega_{\Lambda 0} = 0.685, \qquad \sigma_{8,0} = 0.811. \tag{1}$$

The representative MAAT branch is

$$\Omega_{\mathrm{MAAT},0} = 0.00331, \qquad w_{\mathrm{MAAT}} = -0.801. \tag{2}$$

The MAAT-modified background is modelled by

$$E^2(a) = \Omega_{m0} a^{-3} + \Omega_{\Lambda 0} + \Omega_{\mathrm{MAAT},0} a^{-3(1+w_{\mathrm{MAAT}})}. \tag{3}$$

For the corrected ΛCDM comparison, the reference branch sets $\Omega_{\mathrm{MAAT},0} = 0$ in both the background and the perturbation source. This is important: the ΛCDM baseline is a true reference model, not a MAAT-background model with only the source switched off.

## 3 Modified Growth Equation

We integrate the linear growth equation in the e-fold variable $N = \ln a$:

$$\delta_m'' + (2-q)\delta_m' = \frac{3}{2}\left[\Omega_m(a) + \mu_{\mathrm{MAAT}}(a)\right]\delta_m, \tag{4}$$

where primes denote derivatives with respect to $N$. The deceleration parameter is

$$q(a) = -1 - \frac{a}{2E^2(a)}\frac{\mathrm{d}E^2}{\mathrm{d}a}, \tag{5}$$

and the minimal MAAT source ansatz is

$$\mu_{\mathrm{MAAT}}(a) = \Omega_{\mathrm{MAAT}}(a)(1 - w_{\mathrm{MAAT}}). \tag{6}$$

This source is phenomenological. It is intended as the simplest forward embedding of structural selection into linear growth, not as a final perturbation theory.

The observable growth proxy is computed from

$$f(a) = \frac{\mathrm{d}\ln D}{\mathrm{d}\ln a}, \qquad f\sigma_8(z) = f(z)D(z)\sigma_{8,0}, \tag{7}$$

with $D(z)$ normalized to $D(0) = 1$.

## 4 Selection-Pressure Perturbations

Local MAAT selection pressures are perturbed around their fixed point:

$$\lambda_a(x,t) = \lambda_a^\star + \delta\lambda_a(x,t). \tag{8}$$

In Fourier space, the minimal relaxation-diffusion model is

$$\tau_a \partial_t \delta\lambda_a(k,t) = -\left(1 + D_a k^2\right)\delta\lambda_a(k,t) + S_k(t). \tag{9}$$

The decay rate is therefore

$$\gamma_k = \frac{1 + D_a k^2}{\tau_a}. \tag{10}$$

Since $\tau_a > 0$ and $D_a > 0$, all homogeneous perturbation modes decay:

$$\delta\lambda_a(k,t) = \delta\lambda_a(k,0)e^{-\gamma_k t}. \tag{11}$$

The numerical benchmark uses

$$\tau_a = 1, \qquad D_a = 0.1, \qquad \lambda_a^\star = 0.5. \tag{12}$$

# 5 Numerical Results

## 5.1 Growth-Sector Deviations

The corrected growth comparison gives small deviations over $z < 2$:

$$\max|\Delta D/D| = 0.0591\%, \qquad \max|\Delta f\sigma_8/f\sigma_8| = 0.4570\%. \tag{13}$$

Table 1: Modified growth results for selected redshifts.

| $z$ | $D_{\mathrm{MAAT}}/D_{\Lambda\mathrm{CDM}}$ | $\Delta D/D$ [%] | $\Delta f\sigma_8/f\sigma_8$ [%] |
|---|---|---|---|
| 0.00 | 1.000000 | 0.0000 | 0.4570 |
| 0.30 | 0.999693 | −0.0307 | 0.1717 |
| 0.57 | 0.999557 | −0.0443 | 0.0583 |
| 0.85 | 0.999488 | −0.0512 | 0.0049 |
| 1.00 | 0.999467 | −0.0533 | −0.0103 |
| 2.00 | 0.999409 | −0.0591 | −0.0483 |

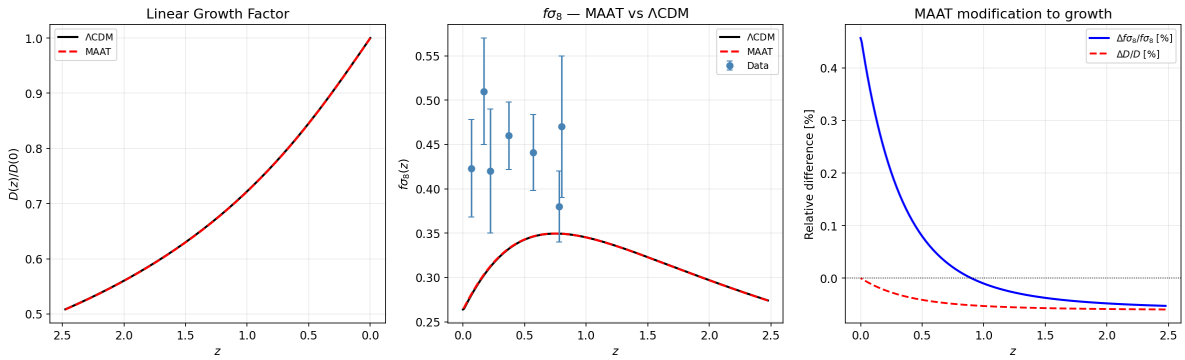


Figure 1: Linear growth embedding. Left: normalized growth factor for ΛCDM and MAAT. Middle: $f\sigma_8(z)$ comparison with representative data points. Right: relative MAAT modification to $D(z)$ and $f\sigma_8(z)$. The modifications remain sub-percent over the tested range.

## 5.2 Selection-Pressure Mode Stability

For $k = \{0.01, 0.10, 0.30, 1.00\}\, h/\mathrm{Mpc}$, the decay rates are

$$\gamma_k = \{1.00001, 1.001, 1.009, 1.1\}.$$

The corresponding decay times are positive and finite:

$$\gamma_k^{-1} = \{0.99999, 0.99900, 0.99108, 0.90909\}.$$

With the toy source $S_k(t)$, the values at $t = 5$ are

$$\delta\lambda_a(t = 5) = \{0.78167, 0.78128, 0.77818, 0.74455\}, \tag{14}$$

showing mild high-$k$ suppression.

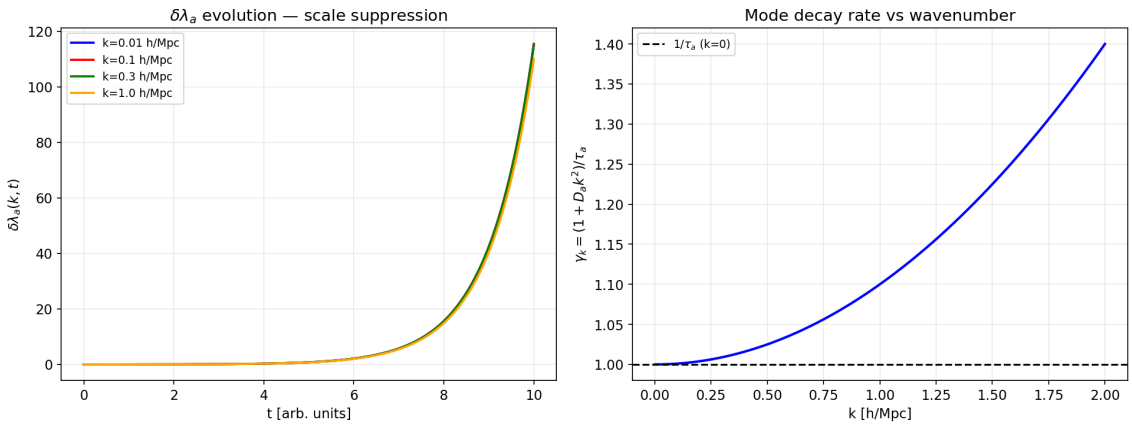


Figure 2: Selection-pressure perturbations. Left: $\delta\lambda_a(k,t)$ under a small growing toy source. Right: mode decay rate $\gamma_k = (1 + D_a k^2)/\tau_a$. Higher-$k$ modes decay faster, giving scale suppression.

### 5.3 Positivity Stress Test

The local field equation

$$\tau_a \partial_t \lambda_a = D_a \nabla^2 \lambda_a - \lambda_a + \lambda_a^\star \tag{15}$$

is tested numerically on a one-dimensional periodic grid with $N = 200$, $T = 20$, and $\Delta t = 0.001$. Three hostile initial conditions are used: near-zero, spike-down, and random-small.

Table 2: Positivity stress test for $\lambda_a(x,t)$.

| **Scenario** | **Initial min.** | **Minimum ever** | **Final min.** |
|---|---|---|---|
| near-zero | 0.010000 | 0.010000 | 0.500000 |
| spike-down | 0.010031 | 0.010031 | 0.500000 |
| random-small | 0.000256 | 0.000256 | 0.500000 |

No negative values occur in the tested scenarios. This is a numerical stress test, not a general theorem for arbitrary nonlinear couplings.

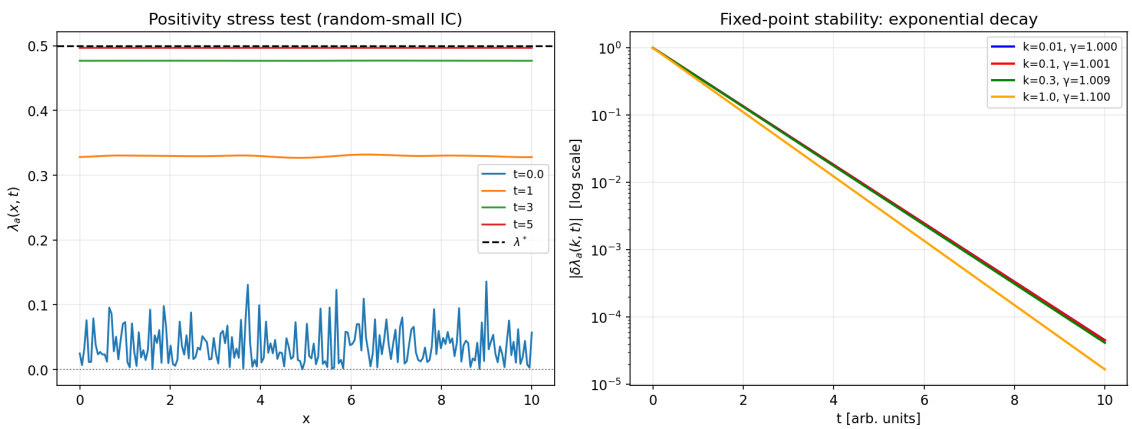


Figure 3: Positivity and fixed-point stability. Left: example positivity evolution for the random-small initial condition. Right: exponential decay of fixed-point perturbations across Fourier modes.

## 6 Energy Conditions

For a normalized effective density $\rho = 1$ and pressure $p = w_{\mathrm{MAAT}}\rho$, the tested branch gives

$$\rho + p = 0.199, \tag{16}$$
$$\rho + 3p = -1.403, \tag{17}$$
$$|p|/\rho = 0.801. \tag{18}$$

Thus:

- the null energy condition is satisfied,
- the weak energy condition is satisfied,
- the dominant energy condition is satisfied,
- the strong energy condition is violated.

The strong-energy-condition violation is expected for an accelerating dark-energy-like component and is not by itself a pathology. The null-energy margin is positive, unlike a pure cosmological constant where $w = -1$ saturates the null-energy condition.

## 7 Interpretation

The core result is:

> *A minimal MAAT growth embedding produces sub-percent growth-sector deviations while preserving linear stability, positivity, and acceptable energy-condition behaviour for the tested branch.*

This resolves the immediate problem raised by Paper 34. Instead of comparing the projection observable directly to $f\sigma_8$, MAAT can be placed into a forward growth equation where its effect is small and well-defined.

The result is not yet a detection claim. It shows viability of the first perturbative embedding.

## 8 Reproducibility

The code and generated artifacts are prepared for the public repository:

https://github.com/Chris4081/structural-selection-principle

Repository folder:

`experiments/maat_growth_perturbation_paper35/`

Run:

```
python3 maat_paper35_reproduction.py
```

The script exports:

- `paper35_summary.json`,
- `paper35_growth_curves.csv`,
- `paper35_dlambda_evolution.csv`,
- `fig_paper35_growth.png`,
- `fig_paper35_perturbation.png`,
- `fig_paper35_positivity.png`.

## 9 Limitations

- The MAAT source term $\mu_{\mathrm{MAAT}}$ is phenomenological.
- No full relativistic perturbation system is solved.
- The comparison is not a likelihood fit to growth data.
- The positivity result is a numerical stress test for selected initial conditions, not a global proof.
- No BAO, supernova, CMB, weak-lensing, or redshift-space distortion likelihood is included.

## 10 Conclusion

This paper provides the first forward growth embedding of MAAT structural selection. Using a true ΛCDM reference and a minimal MAAT source term, the benchmark finds sub-percent growth modifications:

$$\max|\Delta D/D| = 0.0591\%, \qquad \max|\Delta f\sigma_8/f\sigma_8| = 0.4570\%.$$

The local selection-pressure perturbations are stable, positivity is preserved in the tested scenarios, and the energy-condition pattern is consistent with a dark-energy-like effective component.

The next hard step is to replace the phenomenological source ansatz with a full perturbation-level derivation and a multi-probe likelihood analysis.

## References


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